```
namespace messaging
{
    class sender
    {
        queue*q;
    public:
        sender():
            q(nullptr)
        {}

        explicit sender(queue*q_):
            q(q_)
        {}

        template<typename Message>
        void send(Message const& msg)
        {
            if(q)
            {
                q->push(msg);
            }
        }
    };
}
```


Receiving messages is a bit more complicated. Not only do you have to wait for a message from the queue, but you also have to check to see if the type matches any of the message types being waited on and call the appropriate handler function. This all starts with the `receiver` class shown in the following listing.

Listing C.3. The receiver class